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AMITY SCHOOL OF ENGINEERING & TECHNOLOGY

Amity University Uttar Pradesh, Greater Noida

Minor Project (ETMN100)

WEEKLY PROGRESS REPORT (WPR)–8

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Project Title:	WaterTriage: A Data-Driven Water Quality Risk Scoring and Intervention Prioritization System for Uttar Pradesh and Bihar

Conclusion

6.1 Summary of Completed Work

This concluding section consolidates the work documented across WPR-2 through WPR-7 against the objectives registered in the project synopsis. Table 2 maps each declared objective to its delivered outcome as implemented and verified in the WaterTriage system.

Synopsis Objective	Delivered Outcome (as verified in WPR-5–7)
Acquisition and normalization of JJM open water-quality data for UP and Bihar	Achieved — WQMIS public report crawler (politeness-controlled, cached, resumable) delivering 1,245 scored live laboratory reports (UP 687 / Bihar 558), supplemented by the 100,690-event quality-affected habitations registry (2009–2012) and groundwater-quality datasets
Relational database schema (PostgreSQL) for source-level records	Achieved — eight-table normalized schema (states, districts, villages, water_samples, readings, bis_parameters, risk_scores, historical_contamination) with canonical state naming and referential integrity
Contamination severity scoring model per BIS IS 10500	Achieved — weighted per-parameter severity with linear interpolation between acceptable and permissible limits, plus a non-linear acute-hazard escalation invariant (E. coli, arsenic) validated against the live database
Recurrence/trend detection module	Achieved — engine classifying village-parameter pairs as persistent, recurring, watchlist, or no-history by blending registry history with current exceedance; confirmed hotspots include Khokha, Purnia (166 iron events, 2009–2012)
FastAPI backend exposing ranked intervention lists	Achieved — nine endpoints including ranked priority queue, recurrence hotspot ranking, and state comparison; performance optimized from 37 s to 0.58 s via set-based queries
Dashboard for district-wise and state-wise comparison	Achieved — zero-dependency console with priority queue and drill-down verdicts, state comparison, and district-level choropleth maps for both states
Containerized deployment (Docker)	Partially achieved — PostgreSQL services containerized via Docker Compose; backend containerization and Alembic migrations carried into future scope (Section 6.4)

Table 2: Objectives-vs-outcomes mapping for the WaterTriage Minor Project.

6.2 Principal Findings

- **Hydrochemical bifurcation of the study region.** The verified database substantiates a fundamental divide across the Indo-Gangetic plain: Uttar Pradesh exhibits a dual-origin contamination signature (geogenic fluoride averaging 1.73 mg/L with spikes to 12.6 mg/L in Agra, plus agricultural nitrate averaging 49.82 mg/L), whereas Bihar's reducing alluvial aquiers drive pervasive dissolved-iron exceedances (Purnia, Katihar, Madhepura) and geogenic arsenic clustering along the active Ganga meander belt (Buxar, Bhojpur, Bhagalpur). Mean composite risk scores of 14.51 (UP) versus 9.34 (Bihar) reflect these distinct profiles rather than a uniform threat.
- **Validation of the acute-hazard escalation invariant.** The scoring engine's departure from conventional arithmetic WQI formulations was empirically justified: in sample U691097L158S1903 (Lucknow, UP), the arithmetic composite of 24.35 would have falsely classified an E. coli-positive source as Low band; the escalation invariant correctly raised it to Critical. Across the live database, this rule prevented false-negative dilution for every acute pathogen and severe chemical detection.
- **Temporal recurrence as the decisive diagnostic.** Cross-sectional testing alone cannot distinguish transient ingress from chronic aquifer failure. The recurrence engine resolved this: Khokha village (Purnia district, Bihar; Village ID 7568) logged 166 distinct iron exceedances across every observation year 2009–2012 — the single most contaminated habitation in the national historical repository — while central-UP districts Unnao (633 fluoride events) and Firozabad (118 events) exhibit recurrence exceeding 65%. Such chronic zones demand deep confined bores or multi-village piped schemes, not shallow re-boring.
- **Operational translation into intervention strategy.** Composite and recurrence scores were converted into a four-archetype intervention matrix — acute microbial (electro-chlorination; low CapEx), geogenic fluoride (Nalgonda/CSIR-NEERI plants; medium OpEx), dissolved iron (aeration-sedimentation-filtration; very low OpEx), and geogenic arsenic (well capping and transition to piped surface water; high CapEx) — balancing cost profiles against institutional feasibility under the Jal Jeevan Mission.

6.3 Limitations

The conclusions above are bounded by methodological constraints restated from the Discussion. First, the 1,245 live laboratory reports constitute a sample extracted across targeted WQMIS identifier ranges rather than a spatially uniform census; district-level densities reflect crawler coverage, not absolute population vulnerability. Second, a temporal discontinuity exists between the historical registry (2009–2012) and current WQMIS testing cycles (2022–2026), so historical recurrence flags long-term hydrogeological susceptibility rather than real-time contamination. Third, rural-habitation geocoding relies on administrative-hierarchy matching rather than precise GPS coordinates, introducing spatial approximation in choropleth displays. These limitations qualify, but do not negate, the findings.

6.4 Future Scope

Building on the validated prototype, future work will proceed along four lines already identified during the project: (1) spatial kriging and geospatial random-forest interpolation to estimate unmonitored habitations between known test points, addressing coverage gaps; (2) expansion of crawler coverage across mapped identifier ranges toward state-wide scale, with AIKosh dataset integration upon access approval; (3) completion of production hardening — backend containerization, Alembic-managed schema migrations, and the interventions tracking loop with CSV/PDF export of ranked lists for district officials; and (4) longitudinal re-scoring as new WQMIS testing cycles are published, converting the system from a static analysis into a living triage service.

In summary, the WaterTriage system described in the registered synopsis has been implemented, tested (26 automated test cases), and validated against verified real-world data. Its central contribution is an open, independently reproducible triage layer that renders Jal Jeevan Mission water-quality data actionable for district and block-level officials who lack access to the programme's internal analytical infrastructure — demonstrating, within the bounds of an 82-day Minor Project, that data-engineering practice can be applied to a socially consequential public-health problem.